



Heaven's Light is Our Guide

Department of Electrical & Electronic Engineering
Rajshahi University of Engineering & Technology
Rajshahi-6204, Bangladesh

Proforma for submitting Research Project
 Course Code: EEE 3100

Course Title: Electronic Shop Practice

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2.	Name of the Department	:	Electrical & Electronic Engineering
3.	Name of the University	:	Rajshahi University of Engineering & Technology
4.	Title of the Project	:	Smart Mobile Robot with Real-Time Object Detection using TensorFlow Lite.
5.	Brief description of the project: Our project is a dual-mode smart car built on a Raspberry Pi, designed to demonstrate the power of edge AI and robotics. The platform seamlessly switches between two functionalities: Remote-Controlled Object Detection: The car is manually driven via a laptop, using a live camera feed and TensorFlow Lite to identify and classify objects in its path in real-time. Autonomous Line-Following: The car drives itself, using OpenCV-based computer vision to navigate along a designated line on the ground. By combining AI-driven perception with autonomous control, this hybrid system serves as a flexible and low-cost blueprint for real-world applications like smart transport, automated surveillance, and hands-on educational robotics.		
	(a) Objectives and aims of the project: The objective of this project is to develop a dual-mode smart mobile platform that integrates real-time object detection and autonomous line following using Raspberry Pi, OpenCV, and TensorFlow Lite. In object detection mode, the car is remotely controlled via a laptop while identifying and classifying objects through a live camera feed. In line-following mode, it navigates autonomously using computer vision. This hybrid system demonstrates the combination of edge AI and robotic control for flexible navigation, with potential applications in smart transport, surveillance, obstacle-aware mobility, and educational robotics.		
	(b) Review of literature on the subject matter of the project and rationale behind the present initiative: The reality of autonomous vehicles (AVs) is fueled by advancements in AI and computer vision, with real-time object detection at their core. However, a significant barrier remains: ensuring reliable detection in adverse conditions like fog, rain, or poor light. To address this, researchers use sensor fusion, combining cameras with radar and thermal imaging. When paired with deep learning networks (CNNs), this method has boosted accuracy by 19% and reduced false negatives by 22%. Simultaneously, detection algorithms like YOLO and Faster R-CNN have reached remarkable levels of speed and precision. The catch is their high computational cost, demanding powerful hardware like Nvidia Jetson or Raspberry Pi 3s to run effectively. This is where we saw a fascinating gap in the research. The literature is full of examples running powerful AI on powerful machines, but what about bringing that intelligence to the most resource-constrained devices? Almost no one has attempted to run both real-time object		

	detection and autonomous navigation on ultra-light, low-cost hardware. Our initiative boldly steps into this gap. We are building a dual-mode AI-powered mobile platform using a Raspberry Pi Zero 2W a tiny, cost-effective computer with very limited memory and processing power. Our system performs two key tasks: it can be remotely controlled to detect objects using a live video stream, and it can autonomously navigate by following a line on the ground. To achieve this, we manually configured a lightweight framework from the ground up, meticulously optimizing every piece of code to squeeze maximum performance out of the minimal hardware, rather than relying on a generic, pre-built environment. The driving force behind our work is to prove a simple but powerful idea: smart, autonomous technology doesn't need to be expensive or power-hungry. This project not only contributes a novel approach to embedded AI and robotics but also demonstrates a blueprint for affordable smart systems. We envision this technology as an educational tool in classrooms, a foundation for low-cost smart transportation, or a part of remote monitoring systems in developing regions. By keeping the processing on the device itself (edge computing), our system is fast, private, and independent of the cloud.
	(c) Expected results: We expect this project to successfully deliver: 1. A fully operational prototype of a dual-mode smart car capable of switching between remote-controlled object detection and autonomous line-following. 2. Real-time object detection performance, where the system identifies and labels objects in a live video feed with acceptable accuracy and speed for on-the-fly decision-making. 3. Reliable autonomous navigation, with the car consistently and smoothly following a designated line using computer vision.
	(d) Relevance of the project to national development: • Safer Reconnaissance: Send the device into minefields or other hazardous zones to identify objects without risking human lives. • Remote Monitoring: Use it to get a live view of dangerous situations, like a disaster area, from a safe distance. • Low-Cost & Expendable: Perfect for high-risk, one-way missions where losing expensive equipment is not an option.
	(e) Methodology to be adopted in the investigation: Experimental work: • Hardware Core: The project is built around a Raspberry Pi Zero 2 W, which handles all processing. • Visual Input: A 5MP Pi Camera is used to capture the video feed for both operational modes. • Motor Control: An L298N motor driver, connected to the Pi's GPIO pins, controls the motors of the mobile chassis. • Software Environment: A Python virtual environment is set up on the Pi to manage all libraries, primarily OpenCV and TensorFlow Lite.

	• Mode 1 (Object Detection): ○ The car is remotely controlled via keyboard. ○ TensorFlow Lite with a pre-trained MobileNet SSD model is used to detect and classify objects in real-time from the live camera stream. • Mode 2 (Line Following): ○ The car navigates autonomously. ○ OpenCV processes the camera feed to detect and follow a line on the ground. • System Architecture: The entire system is self-contained. All code is in Python and runs directly on the Raspberry Pi without any external processing boards or custom PCBs.					
6.	Financial Statement (item wise):					
	Serial No.	Name of the Equipment's	Specification	Quantity	Unit Price (Taka)	Total Price (Taka)
	1	Raspberry Pi Camera 5MP		1	880	880
	2	Raspberry Pi Zero 2 W		1	4600	4600
	3	Raspberry Pi Zero Camera Adapter		1	250	250
	4	Raspberry Pi Zero Official Case	(with GPIO and Camera Hole)	1	380	380
	5	40 Pin 2.54mm Pitch Male Double Row Straight Pin Header Strip		1	40	40
	6	Aluminum Heat Sink with Adhesive	(for Raspberry Pi 3B/3B+/4B)	1	70	70
	7	Raspberry Pi Power Cable with On/Off Switch	(Micro-USB)	1	190	190
	8	1.5m Mini HDMI to HDMI Cable		1	550	550
	9	Micro SD	32GB	1	450	450
	10	Power Bank		1	2000	2000
	11	Robot Chassis Hawk01	Red	1	332	332
	12	Wheel & Rotary Wheel		2 + 1	145	255
	13	Sanford 3.7V 7000mAh 18650 Battery		2	100	200
	14	L298N H-Bridge Dual Motor Driver		1	183	183
	15	6v 100 RPM DC Gear MOTOR and accessories		2	100	200
					Total Expense (Taka)	10560

DECLARATION/CERTIFICATION

It is certified that-----

- a) The same project has not been submitted to anywhere (UGC/other agency/University) for financial support.
- (b) The research work proposed in this project is not a duplicate work already done or being done in the field (i.e. area of research)
- (c) We agree to accept the terms and conditions developed for the Allocation for Engineering, Science and Technology as mentioned in the Guidelines.
- (d) Project will be provided with access to all available facilities in this university.

Signature of the Course Teacher

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Signature of the Head of the Department



Department of Electrical & Electronic Engineering
Rajshahi University of Engineering & Technology

Project Title	:	Smart Mobile Robot with Real-Time Object Detection using TensorFlow Lite
Project Description	:	This project showcases a low-cost, dual-function smart car built on the Raspberry Pi Zero 2 W, integrating edge AI and robotics. It operates in two modes: object detection via remote control using TensorFlow Lite and live camera feed, and autonomous navigation using OpenCV-based line-following. Designed for real-time performance on a lightweight platform, it demonstrates practical applications in smart transport, surveillance, and educational robotics.
Team Members (Name and Roll No)	:	1. Md. Maruf Haque Shah (2101061) 2. Labib Marwan Hoque (2101064) 3. Al Masrafi Shahria (2101071)

Technical Design and Implementation

The project uses a Raspberry Pi Zero 2 W as the main controller, integrating a 5MP Pi Camera for both object detection and line following. In object detection mode, the car is remotely operated via keyboard input, using TensorFlow Lite with a pretrained MobileNet SSD model for real-time classification. For line following, OpenCV 2 processes the camera feed to track and follow a path autonomously. A L298N motor driver controls the motors on a mobile chassis through GPIO pins. The entire system is programmed in Python and runs fully on the Raspberry Pi without external boards or PCB.

Project Justification and Value

This project addresses the growing need for compact, intelligent robotic platforms capable of performing real-time tasks using edge computing. By combining object detection and autonomous navigation, it demonstrates how AI and robotics can be integrated for practical applications like smart transport, surveillance, and educational tools. Using pretrained models and low-cost hardware, the system offers a scalable and efficient solution for environments where cloud-based processing is not feasible. The project development follows clear stages: hardware setup, software integration, vision-based tracking, and real-time inference. Design choices prioritize cost-effectiveness, modularity, and real-world relevance, making it technically impactful and socially beneficial.

Project Planning and Team Roles

Our team of three planned and built this project collaboratively, working together through each stage from assembling the hardware to coding and testing the system. Instead of rigid role divisions, we shared responsibilities and helped each other wherever needed. One focused more on motor control and wiring,

another handled the computer vision and AI integration, while the third worked on system debugging and coordination. All decisions technical and economic were discussed as a group to keep the project simple, efficient, and cost-effective. We met regularly, shared ideas openly, and made sure everyone stayed involved throughout the process.

(Addresses Complex Engineering Problem criteria)

The project tackles a real-world challenge: how to run both object detection and autonomous navigation on a small, low-power device like the Raspberry Pi Zero 2 W. It brings together different fields—computer vision, robotics, AI, and embedded systems—into one compact platform. The main complexity lies in processing camera input, making decisions in real time, and controlling the motors smoothly, all with limited hardware resources. We approached this by breaking the problem down into manageable parts, using lightweight algorithms, and carefully optimizing our system to make everything work together efficiently.

Smart Mobile Robot with Real-Time Object Detection using TensorFlow Lite

Poster Presentation Abstract

Background & motivation.

Small, low-cost robots usually trade capability for speed. We set out to show you can have both on a tiny edge device. Our robot runs entirely on a Raspberry Pi Zero 2 W—no companion boards and no cloud—and can both recognize objects from a live camera stream and follow a path on its own. The aim is a platform that classrooms, hobbyists, and labs can afford, rebuild, and extend.

System overview.

A single 5 MP Pi Camera and Python codebase support two modes:

1. **Remote object detection.** The car is keyboard-driven from a host while on-board TensorFlow Lite (MobileNet-SSD) provides real-time labels and boxes, helping the operator steer toward targets without any network link.
2. **Autonomous line following.** With OpenCV, each frame is thresholded to isolate the track, a midline is extracted, and a proportional controller converts lateral error into left/right wheel speeds.

An **L298N** driver on the Pi's GPIO powers a two-wheel chassis. Vision, inference, and motor control all execute on the Pi Zero 2 W to keep wiring simple and latency low.

Why these choices.

- **Edge-only processing** removes network lag and works in constrained settings.
- **Lightweight vision**—TFLite + MobileNet-SSD and classic OpenCV—fits the Pi's compute budget while staying interpretable.
- **Simple hardware**—Pi Camera + L298N—keeps costs down and replication easy.
- **One language (Python)** speeds iteration and lowers the learning curve.

Implementation snapshot.

- **Detection:** Camera frames are resized and normalized for the TFLite interpreter; outputs are post-processed to draw labels and confidences. Keyboard commands map to GPIO for differential drive.
- **Line following:** A region of interest is processed to find the line centroid or fit a contour; smoothed error feeds a P-controller to reduce oscillation.
- **Motor control:** A small GPIO layer sets direction and PWM with safety clamps (min/max duty, stop-on-fault).
- **Runtime:** Both modes share the same camera stream and event loop; switching modes swaps only the processing callback.

Engineering challenges tackled.

The core difficulty is doing perception and control in real time on limited hardware. We balanced inference load with control-loop timing, trimmed pre-/post-processing, and tuned steering to stay stable without extra sensors. The system was built as small, testable pieces—camera I/O,

preprocessing, inference, control, actuation—then profiled and tightened until end-to-end behavior was smooth.

Team process.

Three of us built the system collaboratively. One leaned into motor control and wiring, one into computer vision and AI, and one into integration and debugging. We made technical and economic decisions together, kept the design lean, and iterated with short check-ins and hands-on tests.

Value & applications.

The robot shows how edge AI and classical robotics meet in a compact package. It suits **smart transport** demos (lane keeping with target cues), **inspection/surveillance** where bandwidth is scarce (local detection, no data egress), and **education**, letting learners see the path from pixels to motion on hardware they can assemble themselves. Using pretrained models and commodity parts makes the platform **replicable** and **budget-friendly**.

Limitations & ethics.

Staying on-device limits model size and headroom. Lighting changes can disrupt line segmentation; adaptive thresholds help but don't solve every case. We rely on general MobileNet-SSD classes and avoid collecting personal data; we emphasize responsible, context-appropriate use in demos and notes.

Results & demonstration.

The poster highlights (a) live, labeled detections during remote operation and (b) autonomous laps on a taped track—both using the same camera and processor. We include wiring from GPIO to L298N, the code layout, and timing for the perception-control loop to show real-time performance and clean switching between modes.

Future work.

Next steps include task-specific detector tuning, obstacle avoidance and stop-line handling, adding a low-cost IMU or ultrasonic sensor, and testing INT8/quantized models for extra speed. A small web dashboard could replace keyboard teleop, and a teaching pack could help instructors adopt the platform.

One-line contribution.

A fully on-Pi, dual-mode robot that pairs TensorFlow Lite detection with OpenCV line following to deliver practical, real-time performance on very low-cost hardware.